

Nonlinear wave equation for an elastic string: derivation, symmetries, conserved quantities

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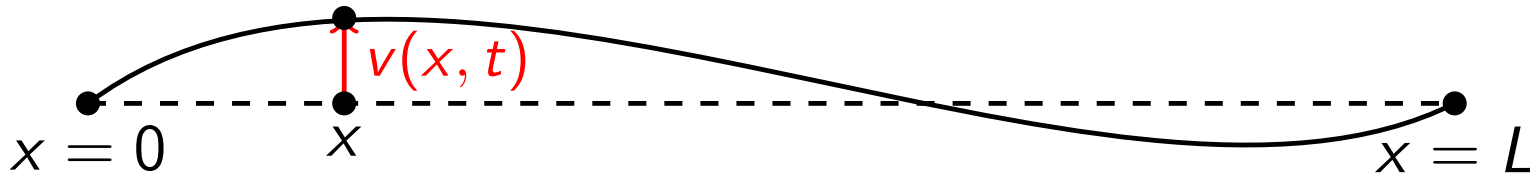
Linear Waves

Linear waves in 1D elastic media have several important physical applications:

1. vibrations of an elastic string e.g. violin or piano
2. sound waves e.g. pipe organ or flute
3. seismic waves e.g. earth tremor

$v(x, t)$ = transverse displacement

equilibrium is $v(x, t) \equiv 0$



Governing equation for linear waves

$$v_{tt} - c^2 v_{xx} = 0, \quad c = \text{const.}$$

Physical conditions under which the linear wave equation holds for the motion of an elastic string:

1. constant tension
2. small angle approximation ($|v_x| \ll 1$)
3. no longitudinal motion

For motion that involves *large deformations* from equilibrium,
all three conditions no longer hold.

Waves in any real elastic material will exhibit some *longitudinal motion* (i.e. compression/rarefaction will occur).

*We want to study correct physical wave equation
for an elastic string without restrictions.*

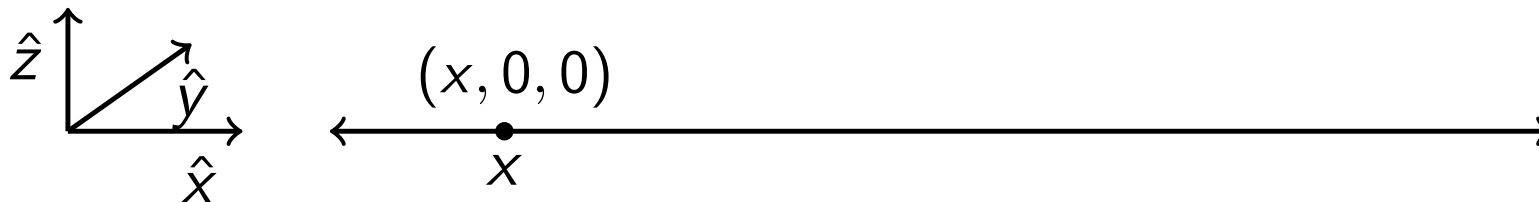
Nonlinear wave equation for 1D elastic string

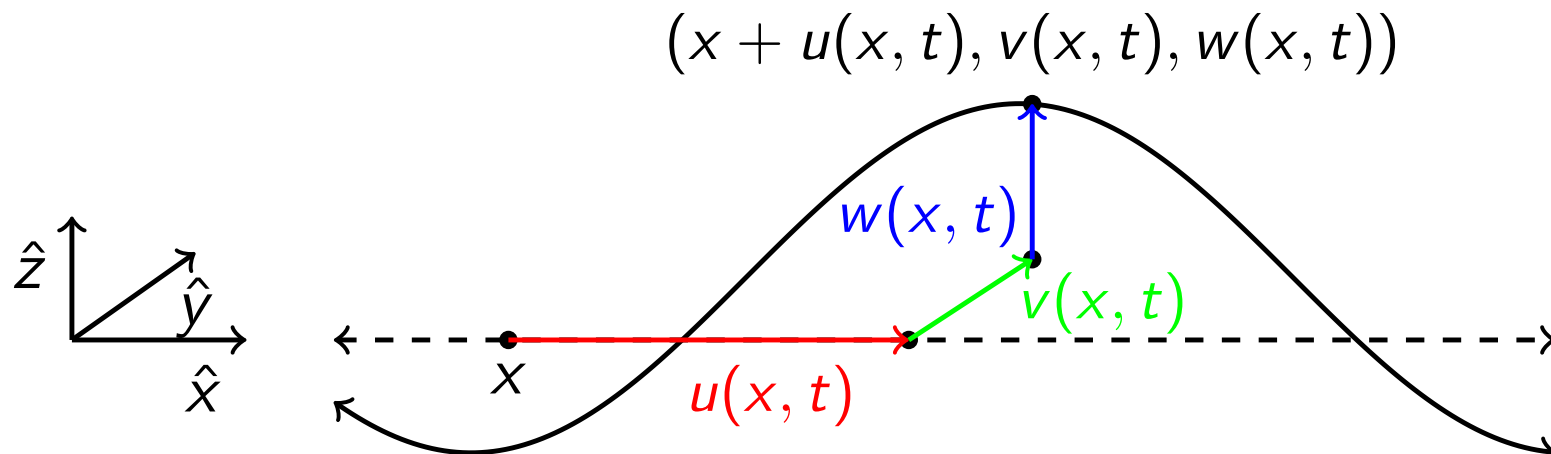
- dynamical state of an elastic string can involve motion in both transverse directions and longitudinal direction
- physical elastic string will have finite length with boundary conditions at the ends
- in a very long elastic string, motion in interior does not depend on boundary conditions

For mathematical simplicity:

consider only free motion without boundary conditions

$v, w =$ transverse displacements; $u =$ longitudinal displacement
equilibrium is $v, w \equiv 0$ and $u \equiv 0$





Elastic string wave equations

$$u_{tt} = c^2 u_{xx} + (\tilde{c}^2 - c^2) \left((1 + u_x) / \sqrt{(1 + u_x)^2 + v_x^2 + w_x^2} \right)_x$$

$$v_{tt} = c^2 v_{xx} + (\tilde{c}^2 - c^2) \left(v_x / \sqrt{(1 + u_x)^2 + v_x^2 + w_x^2} \right)_x$$

$$w_{tt} = c^2 w_{xx} + (\tilde{c}^2 - c^2) \left(w_x / \sqrt{(1 + u_x)^2 + v_x^2 + w_x^2} \right)_x$$

where $c, \tilde{c} = \text{const.} > 0$

Survey of literature on derivations

First derivation: W.F. Osgood 1925, “Advanced Calculus” (book)

Widely-cited paper: G.F. Carrier 1945, Q. J. Appl. Math.

- applied Newton’s law framework
- restricted motion to a plane
- assumes linear stress-strain relation (Hooke’s law)

Longitudinal dynamical variable u and transverse dynamical variable v
 $\Longleftrightarrow w = 0$ in the nonlinear wave equations

Stress $\sigma = E\lambda$, where $\lambda = \sqrt{(1 + u_x)^2 + v_x^2} - 1$ is strain (net stretch)
and E is elastic modulus (constant)

Dynamical tension $T = T_0 + A\sigma$, where T_0 is tension in equilibrium
and A is cross-section area (constant)

$$\rho A u_{tt} = \left((1 + u_x) T / (\lambda + 1) \right)_x, \quad \rho A v_{tt} = \left(v_x T / (\lambda + 1) \right)_x$$

where ρ is density (constant)

Subsequent major paper: R. Narasimha 1968, J. Sound and Vibration

Similarly to Osgood/Carrier

- applied Newton's law framework
- assumes linear stress-strain relation (Hooke's law)

Differences

- used Lagrangian variables (position and time for motion of particles comprising the elastic string)
- *no restriction* on transverse motion
- emphasizes that, in general, *longitudinal mode is excited* ($u \neq 0$)
- points out that *transverse motion in a plane may be unstable*

When the equations are re-written in the “lab” variables x and t , they are equivalent to the equations for (u, v, w)

Nonlinear elasticity approach: S. Antman 2005, “Nonlinear Problems of Elasticity” (book)

- uses conservation of momentum and angular momentum in vector variables
- no restriction on transverse motion
- considers general *nonlinear stress-strain* relation

position vector of material points $\vec{r}(t, x) = (x + u, v, w)$

elastic force vector $\vec{F}^e$

Conservation of momentum states

$$\rho A \vec{r}_{tt} = \vec{F}_x^e$$

Conservation of angular momentum leads to the condition $\vec{F}^e \times \vec{r}_x = 0$ (called “perfect flexibility”)

$\implies$

$$\vec{F}^e = K(\lambda) \widehat{r}_x$$

where $\widehat{r}_x = |\vec{r}_x|^{-1} \vec{r}_x$ is unit tangent vector, and $K(\lambda)$ is called the stress-strain function.

Stress-strain function $K(\lambda)$ must satisfy two physical conditions

- K increases with λ : string lengthens only with increasing K
- $K \rightarrow \infty$ as $\lambda \rightarrow \infty$: an infinitely large stretch occurs only by an infinite tension

Equation of motion for $\vec{r}(t, x)$ has explicit component form

$$u_{tt} = \left(k(\lambda)(1 + u_x) \right)_x$$

$$v_{tt} = \left(k(\lambda)v_x \right)_x$$

$$w_{tt} = \left(k(\lambda)w_x \right)_x$$

with

$$k(\lambda) = \frac{K(\lambda)}{(\lambda + 1)\rho A}, \quad \lambda = \sqrt{(1 + u_x)^2 + v_x^2 + w_x^2} - 1$$

- Linear function $K(\lambda) = T_0 + AE\lambda$ yields Narasimha's equations for elastic string
- Nonlinear functions $K(\lambda)$ describe nonlinearly elastic string (i.e. nonlinear tension force)

Continuum mechanics framework: Serrano et al. 2019, *Internat. J. Solids and Structures*

- uses *action principle* in terms of displacement variables
- allows for general transverse motion
- based on stored potential energy which is a function of strain tensor
- equations of motion given by variational derivative of action principle

equilibrium is $\vec{X} = (x, 0, 0)$ (called “reference configuration”)

$\vec{r} = (x + u, v, w)$ position of material elements comprising string (called “current configuration”)

displacement is $\vec{R} = \vec{r} - \vec{X} = (u, v, w)$

kinetic energy of material element $\frac{1}{2}\rho A |\vec{R}_t|^2 dx$

a material element $d\vec{X}$ is stretched to $d\vec{r} = \mathbf{F} \cdot d\vec{X}$, where $\mathbf{F} = \partial_{\vec{X}} \vec{r}$ is the deformation gradient tensor

stored potential energy of a material element is $W(\text{strain tensor})dx$, where the strain tensor, quantifying the deformation, vanishes when $\mathbf{F} \in \text{SO}(3)$

some obvious choices for the strain tensor are $(\mathbf{F}^\top \mathbf{F})^p - \mathbf{I} = (\mathbf{F}^\top \mathbf{F}) \cdots (\mathbf{F}^\top \mathbf{F}) - \mathbf{I}, p \neq 0$, and $\ln \mathbf{F}^\top \mathbf{F}$

physically, $\mathbf{F}^\top \mathbf{F}$ gives the norm of the stretched material element, $d\vec{r} \cdot d\vec{r} = (d\vec{X})^\top \mathbf{F}^\top \mathbf{F} d\vec{X}$

for strings, $\mathbf{F} = |\vec{r}_x|(\widehat{r}_x \otimes \widehat{i})$ and $\mathbf{F}^\top \mathbf{F} = |\vec{r}_x|^2(\widehat{i} \otimes \widehat{i}) = (1 + \lambda)^2(\widehat{i} \otimes \widehat{i})$, where $\lambda = |\vec{r}_x| - 1 = \sqrt{(1 + u_x)^2 + v_x^2 + w_x^2} - 1$ is the strain (net stretch)

effectively, the stored potential energy can be defined to be $W(\lambda)dx$

Action principle is total kinetic energy minus total potential energy

$$S = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left(\frac{1}{2} \rho A |\vec{R}_t|^2 - W(\lambda) \right) dx dt$$

$\implies$ variational derivative yields (material) equations of motion

$$\frac{\delta S}{\delta \vec{R}} = -\rho A \vec{R}_{tt} + \left((W'(\lambda)/(\lambda + 1)) \vec{r}_x \right)_x = 0$$

$\implies$ nonlinear elastic string equations

$$\vec{R}_{tt} = (k(\lambda) \vec{r}_x)_x, \quad k(\lambda) = \frac{W'(\lambda)}{(\lambda + 1) \rho A}, \quad \lambda = |\vec{r}_x| - 1, \quad \vec{r} = x \hat{i} + \vec{R}$$

where $W'(\lambda) = K(\lambda)$ is the stress-strain function.

- Potential energy is typically a quadratic function of strain

$$W(\lambda) = \frac{1}{2} c^2 (\lambda + 1)^2 + (\tilde{c}^2 - c^2) \lambda$$

$\implies$ linear stress-strain $K(\lambda) = c^2 \lambda + \tilde{c}^2$

- Reduces to the Carrier/Narasimha equations for $\vec{R} = (u, v, w)$

Lagrangian and Hamiltonian of elastic string equations

- Lagrangian

$$L[\vec{R}] = \frac{1}{2}\rho A |\vec{R}_t|^2 - W(\lambda)$$

$$\frac{\delta L}{\delta \vec{R}} = -\rho A \vec{R}_{tt} + \underbrace{\left((W'(\lambda)/(\lambda + 1)) \vec{r}_x \right)}_{\rho A k(\lambda)} = 0$$

- Hamiltonian, H , can be obtained from L via a Legendre transform using the momentum

$$\vec{p} = \frac{\partial L}{\partial \vec{R}_t} = \rho A \vec{R}_t$$

$\implies$

$$H[\vec{p}, \vec{R}] = \vec{p} \cdot \vec{R}_t - L = \frac{|\vec{p}|^2}{2\rho A} + W(\lambda)$$

Hamilton's equations of motion:

$$\vec{R}_t = \frac{\delta H}{\delta \vec{p}} = \frac{1}{\rho A} \vec{p} \quad \text{and} \quad \vec{p}_t = -\frac{\delta H}{\delta \vec{R}} = \rho A \left(k(\lambda) \vec{r}_x \right)_x$$

- These equations are evolution PDEs which have a unique solution for given initial data $(u, v, w)|_{t=0}$ and $(u_t, v_t, w_t)|_{t=0}$.

Point symmetries

- Transformations acting in the space of kinematic variables (x, t, u, v, w) under which the form of the wave equations is preserved
- The wave equations manifest some evident point symmetries:
time and space translations; rotations in the transverse (v, w) plane

Goal: Find all non-manifest point symmetries

Consider one-parameter group of transformations

$$\begin{aligned}x &\rightarrow \tilde{x} = x + \epsilon \xi(x, t, u, v, w) + O(\epsilon^2) \\t &\rightarrow \tilde{t} = t + \epsilon \tau(x, t, u, v, w) + O(\epsilon^2) \\u &\rightarrow \tilde{u} = u + \epsilon \eta^u(x, t, u, v, w) + O(\epsilon^2) \\v &\rightarrow \tilde{v} = v + \epsilon \eta^v(x, t, u, v, w) + O(\epsilon^2) \\w &\rightarrow \tilde{w} = w + \epsilon \eta^w(x, t, u, v, w) + O(\epsilon^2)\end{aligned}$$

where $\epsilon = 0$ gives the identity transformation.

- Need to know how transformations act on functions of (t, x) , because the string equations contain derivatives of (u, v, w)

- Simplest way is by working with infinitesimal transformations

$$\begin{aligned}(u(t, x), v(t, x), w(t, x)) &\rightarrow (\tilde{u}(t, x), \tilde{v}(t, x), \tilde{w}(t, x)) \\ &= (u(t, x), v(t, x), w(t, x)) \\ &\quad + \epsilon (P^u(t, x), P^v(t, x), P^w(t, x)) + O(\epsilon^2)\end{aligned}$$

which is given by

$$P^u = \eta^u - \xi u_x - \tau u_t \equiv \delta u, \quad P^v = \eta^v - \xi v_x - \tau v_t \equiv \delta v, \quad P^w = \eta^w - \xi w_x - \tau w_t \equiv \delta w$$

Bluman & Anco 2002, "Symmetry and Integration Methods for Differential Equations" (book)

- δ denotes infinitesimal change given by the ϵ -derivative at $\epsilon = 0$

Example: space translation $x \rightarrow \tilde{x} = x + \epsilon$ has $\xi = 1$, $\tau = 0$, $\eta = 0$

Transformation on variables gives

$$\begin{aligned}\tilde{u} &= u && \text{since no change in } u \\ &= \tilde{u}(\tilde{x}, \tilde{t}) = u(x, t) = u(\tilde{x} - \epsilon, \tilde{t}) && \text{by } x = \tilde{x} - \epsilon \text{ and } t = \tilde{t}\end{aligned}$$

$\implies u(x, t) \rightarrow \tilde{u}(x, t) = u(x - \epsilon, t)$ equivalent transformation on functions

Infinitesimal change in $u(t, x)$ is given by $P^u = \left. \frac{\partial u(x - \epsilon, t)}{\partial \epsilon} \right|_{\epsilon=0} = -u_x$

How to find infinitesimal point symmetries

Preservation of the form of the elastic string equations

$$\vec{R}_{tt} = (k(\lambda)(\hat{i} + \vec{R}_x))_x$$

under a symmetry means that the symmetry transformation maps solutions into solutions

$\implies$ Infinitesimal change in the form of the equations must vanish for all solutions:

$$\delta \left(\vec{R}_{tt} - (k(\lambda)(\hat{i} + \vec{R}_x))_x \right) = 0 \quad \text{when} \quad \vec{R}_{tt} = (k(\lambda)(\hat{i} + \vec{R}_x))_x$$

where $\delta \vec{R} = \vec{P} = (P^u, P^v, P^w)$.

Substitute $\delta \vec{R}_{tt} = \partial_t^2 \vec{P}$ and $\delta \vec{R}_x = \partial_x \vec{P}$ and $\delta k(\lambda) = \frac{k'(\lambda)}{\lambda + 1} (\hat{i} + \vec{R}_x) \cdot \partial_x \vec{P}$

where ∂ denotes the derivative acting by chain rule

$\implies$

$$\partial_t^2 \vec{P} - \partial_x \left(\frac{k'(\lambda)}{\lambda + 1} ((\hat{i} + \vec{R}_x) \cdot \partial_x \vec{P}) (\hat{i} + \vec{R}_x) + k(\lambda) \partial_x \vec{P} \right) = 0$$

$$\text{when} \quad \vec{R}_{tt} = (k(\lambda)(\hat{i} + \vec{R}_x))_x$$

called *symmetry determining equations*

More compact tensorial expression for the determining equations

$$\partial_t^2 \vec{P} \Big|_{\vec{R}_{tt} = (k(\lambda)(\hat{i} + \vec{R}_x))_x} = \partial_x (\mathbf{K} \cdot \partial_x \vec{P})$$

where

$$\mathbf{K} = \frac{k'(\lambda)}{\lambda + 1} (\hat{i} + \vec{R}_x) \odot (\hat{i} + \vec{R}_x) + k(\lambda) \mathbf{I}$$

- For point symmetries

$$\vec{P} = \vec{\eta} - \xi \vec{R}_x - \tau \vec{R}_t$$

where $\vec{\eta} = (\eta^u, \eta^v, \eta^w)$, ξ , τ are unknown functions of t, x , $\vec{R} = (u, v, w)$

- By inspection:

time translation $\tau = 1$, $\xi = 0$, $\vec{\eta} = 0$:

$$\tilde{t} = t + \epsilon$$

space translation $\xi = 1$, $\tau = 0$, $\vec{\eta} = 0$:

$$\tilde{x} = x + \epsilon$$

transverse rotation $\vec{\eta} = (0, w, -v)$, $\xi = 0$, $\tau = 0$:

$$\begin{pmatrix} \tilde{v} \\ \tilde{w} \end{pmatrix} = \begin{pmatrix} \cos(\epsilon) & \sin(\epsilon) \\ -\sin(\epsilon) & \cos(\epsilon) \end{pmatrix} \begin{pmatrix} v \\ w \end{pmatrix}$$

constant shifts in (u, v, w)

Maple computation to find the point symmetries

Main steps:

1. Set up the general form for the elastic string equations and the general form for a point symmetry.
2. Compute the determining equations for point symmetries.
3. Bring the determining equations to simplest form such that linear combinations and cross-differentiations do not yield any equations of lower order in derivatives.

Different cases may arise for the resulting simplified equations.

4. Solve in each case the system of equations for the unknowns.

(Show worksheet)

List of solutions modulo translations and transverse rotations

- Maple computation yields a classification of all point symmetries
- Comprises three separate cases depending on the form of $k(\lambda)$

1. Arbitrary $k(\lambda)$

6 point symmetries

- dilation*: $(\tilde{x}, \tilde{t}, \tilde{u}, \tilde{v}, \tilde{w}) = e^\epsilon(x, t, u, v, w)$
- time-dependent shift in u* : $\tilde{u} = u + \epsilon t$
- time-dependent shift in v* : $\tilde{v} = v + \epsilon t$
- time-dependent shift in w* : $\tilde{w} = w + \epsilon t$
- rotations in the $(x + u, v)$ plane*:
$$\begin{pmatrix} \tilde{x} + \tilde{u} \\ \tilde{v} \end{pmatrix} = \begin{pmatrix} \cos(\epsilon) & \sin(\epsilon) \\ -\sin(\epsilon) & \cos(\epsilon) \end{pmatrix} \begin{pmatrix} x + u \\ v \end{pmatrix}$$
- rotations in the $(x + u, w)$ plane*:
$$\begin{pmatrix} \tilde{x} + \tilde{u} \\ \tilde{w} \end{pmatrix} = \begin{pmatrix} \cos(\epsilon) & \sin(\epsilon) \\ -\sin(\epsilon) & \cos(\epsilon) \end{pmatrix} \begin{pmatrix} x + u \\ w \end{pmatrix}$$

2. $k(\lambda) = k_0(1 + \lambda)^p$, $p \neq -4, -1, 0$, $k_0 = \text{const.} \neq 0$

1 extra point symmetry

i. *scaling*: $\tilde{t} = e^\epsilon t$, $\tilde{u} = e^{-\frac{2}{p}\epsilon}(x + u) - x$, $\tilde{v} = e^{-\frac{2}{p}\epsilon}v$, $\tilde{w} = e^{-\frac{2}{p}\epsilon}w$

invariants are x , $t^{\frac{2}{p}}v$, $t^{\frac{2}{p}}w$, $t^{\frac{2}{p}}(x + u)$

3. $k(\lambda) = k_0(1 + \lambda)^{-4}$, $k_0 = \text{const.} \neq 0$

1 extra point symmetry

i. *time-dependent (conformal) scaling*:

$$\tilde{t} = \frac{t}{1 - \epsilon t}, \quad \tilde{u} = \frac{u + \epsilon x t}{1 - \epsilon t}, \quad \tilde{v} = \frac{v}{1 - \epsilon t}, \quad \tilde{w} = \frac{w}{1 - \epsilon t}$$

invariants are x , $\frac{v}{t}$, $\frac{w}{t}$, $\frac{x + u}{t}$

Physical meaning of the $k(\lambda)$ in each case

The stress-strain function, $K(\lambda)$, in each case can be found using $k(\lambda) = \frac{K(\lambda)}{(\lambda+1)\rho A}$:

1. Arbitrary $k(\lambda) \implies$ arbitrary $K(\lambda)$
2. $k(\lambda) = k_0(1 + \lambda)^p$, $p \neq -4, -1, 0$, $k_0 = \text{const.} \neq 0$
 $\implies K(\lambda) = K_0(1 + \lambda)^{p+1}$, $K_0 = \text{const.} \neq 0$
3. $k(\lambda) = k_0(1 + \lambda)^{-4}$, $k_0 = \text{const.} \neq 0$
 $\implies K(\lambda) = K_0(1 + \lambda)^{-3}$, $K_0 = \text{const.} \neq 0$

Conservation laws

- By applying Noether's theorem [Bluman, Cheviakov, & Anco 2010, "Applications of symmetry methods to partial differential equations" \(book\)](#), we can use the infinitesimal symmetries $\delta \vec{R} = \vec{P}$ to find the conservation laws admitted by the elastic string equations.
- *Variational symmetries* are transformations that preserve the action. Equivalently, they change the Lagrangian L by a total space-time derivative,

$$\delta_{\vec{P}} L \equiv \partial_t A + \partial_x B$$

where A and B are functions of the dynamical variables and their derivatives.

- The *variational derivative annihilates total space-time derivatives*. Using this property, we can find the variational symmetries by checking the condition

$$\frac{\delta(\delta_{\vec{P}} L)}{\delta \vec{R}} = 0$$

which holds off of solutions.

- For any variation $\delta \vec{R} = \vec{P}$, the infinitesimal change in the Lagrangian is

$$\begin{aligned}
\delta_{\vec{P}} L &= \partial_t \vec{P} \cdot \partial_{\vec{R}_t} L + \partial_x \vec{P} \cdot \partial_{\vec{R}_x} L \\
&= \partial_t (\vec{P} \cdot \partial_{\vec{R}_t} L) - \vec{P} \cdot \partial_t \partial_{\vec{R}_t} L + \partial_x (\vec{P} \cdot \partial_{\vec{R}_x} L) - \vec{P} \cdot \partial_x \partial_{\vec{R}_x} L \\
&= \partial_t (\vec{P} \cdot \partial_{\vec{R}_t} L) + \partial_x (\vec{P} \cdot \partial_{\vec{R}_x} L) - \vec{P} \cdot \frac{\delta L}{\delta \vec{R}}
\end{aligned}$$

Rearranging this expression, we obtain

$$\begin{aligned}
\vec{P} \cdot \frac{\delta L}{\delta \vec{R}} &= \partial_t (\vec{P} \cdot \partial_{\vec{R}_t} L) + \partial_x (\vec{P} \cdot \partial_{\vec{R}_x} L) - \delta_{\vec{P}} L \\
&= \partial_t (\vec{P} \cdot \partial_{\vec{R}_t} L - A) + \partial_x (\vec{P} \cdot \partial_{\vec{R}_x} L - B)
\end{aligned}$$

for variational symmetries.

- When equation evaluated on solutions, $\frac{\delta L}{\delta \vec{R}} = 0$, the space-time total derivative on the right-hand side evaluates to zero, yielding a conservation law:

$$\vec{P} \cdot \frac{\delta L}{\delta \vec{R}} = \partial_t T + \partial_x X = 0$$

where

$$T = \vec{P} \cdot \partial_{\vec{R}_t} L - A$$

is the *conserved density*, and

$$X = \vec{P} \cdot \partial_{\vec{R}_x} L - B$$

is its *flux*.

- To find T and X explicitly, we repeatedly integrate by parts in $\vec{P} \cdot \frac{\delta L}{\delta \vec{R}}$ until the expression takes the form of a space-time total derivative.

Maple computation to find the conservation laws

The main steps for computing conservation laws in Maple are:

1. In each case, determine the conditions on the symmetry generator such that the variational derivative annihilates the dot product of the symmetry generator and the string equations.
2. Substitute these conditions into the symmetry generator.
3. Integrate by parts in the dot product of the modified symmetry generator and the string equations to obtain the explicit expressions for the conserved density and its flux.

Variational symmetries

- Case 1: Arbitrary $k(\lambda)$

In this case, there are 12 symmetries, of which 11 are variational symmetries. The uniform scaling is not variational.

- Case 2: $k(\lambda) = k_0(1 + \lambda)^p$, $p \neq -4, -1, 0$, $k_0 = \text{const.} \neq 0$

In this case, there are 13 symmetries, of which 12 are variational. A combination of the scaling symmetries given by

$$\vec{P} = \left(\frac{1}{p+4}(-(3p+4)tu_t + p(x+u)) - x(1+u_x), \right. \\ \left. \frac{1}{p+4}(-(3p+4)tv_t + pv) - xv_x, \right. \\ \left. \frac{1}{p+4}(-(3p+4)tw_t + pw) - xw_x \right)$$

is a variational symmetry.

- Case 3: $k(\lambda) = k_0(1 + \lambda)^{-4}$, $k_0 = \text{const.} \neq 0$

In this case, there are 14 symmetries, of which 13 are variational. The uniform scaling symmetry is not variational.

Conserved densities and their fluxes

The variational symmetries in each case gives the following conservation laws. The physical meaning of the conserved densities $T_6, T_7, T_8, T_{12}, T_{13}$, and T_{14} is not known.

- Case 1: Arbitrary $k(\lambda)$

In this case, there are 11 conservation laws corresponding to the 11 variational symmetries:

i. time translation: $\vec{P} = (-u_t, -v_t, -w_t)$ gives

$$T_1 = \frac{1}{2}\rho A(u_t^2 + v_t^2 + w_t^2) + W(\lambda) = H,$$
$$X_1 = -\frac{W'(\lambda)}{\lambda + 1}((1 + u_x)u_t + v_x v_t + w_x w_t)$$

where T_1 is the *energy* density.

ii. space translation: $\vec{P} = (-u_x, -v_x, -w_x)$ gives

$$T_2 = \rho A(u_t u_x + v_t v_x + w_t w_x),$$
$$X_2 = -\frac{1}{2}\rho A(u_t^2 + v_t^2 + w_t^2) + W(\lambda)$$
$$- \frac{W'(\lambda)}{\lambda + 1}((1 + u_x)u_x + v_x^2 + w_x^2)$$

iii. constant shift in u : $\vec{P} = (1, 0, 0)$ gives

$$T_3 = u_t, \quad X_3 = -k(\lambda)(1 + u_x)$$

where T_3 is the *x-momentum* density.

iv. constant shift in v : $\vec{P} = (0, 1, 0)$ gives

$$T_4 = v_t, \quad X_4 = -k(\lambda)v_x$$

where T_4 is the *y-momentum* density.

v. constant shift in w : $\vec{P} = (0, 0, 1)$ gives

$$T_5 = w_t, \quad X_5 = -k(\lambda)w_x$$

where T_5 is the *z-momentum* density.

vi. time dependent shift in u : $\vec{P} = (t, 0, 0)$ gives

$$T_6 = u - tu_t, \quad X_6 = tk(\lambda)(1 + u_x)$$

vii. time dependent shift in v : $\vec{P} = (0, t, 0)$ gives

$$T_7 = v - tv_t, \quad X_7 = tk(\lambda)v_x$$

viii. time dependent shift in w : $\vec{P} = (0, 0, t)$ gives

$$T_8 = w - tw_t, \quad X_8 = tk(\lambda)w_x$$

ix. rotations in the (v, w) plane: $\vec{P} = (0, -w, v)$ gives

$$T_9 = vw_t - v_tw, \quad X_9 = k(\lambda)(v_xw - vw_x)$$

where T_9 is the *angular momentum density in the x -direction*.

x. rotations in the $(x + u, w)$ plane: $\vec{P} = (w, 0, -(x + u))$ gives

$$T_{10} = u_tw - (x + u)w_t, \quad X_{10} = k(\lambda)((x + u)w_x - (1 + u_x)w)$$

where T_{10} is the *angular momentum density in the y -direction*.

xi. rotations in the $(x + u, v)$ plane: $\vec{P} = (-v, x + u, 0)$ gives

$$T_{11} = (x + u)v_t - u_tv, \quad X_{11} = k(\lambda)((1 + u_x)v - (x + u)v_x)$$

where T_{11} is the *angular momentum density in the z -direction*.

- Case 2: $k(\lambda) = k_0(1 + \lambda)^p$, $p \neq -4, -1, 0$, $k_0 = \text{const.} \neq 0$

In this case, corresponding to the 12 variational symmetries, there are 12 conservation laws, including specializations of the 11 from Case 1 plus one extra:

- i. scaling: $\vec{P} = (\frac{1}{p+4}(-(3p+4)tu_t + p(x+u)) - x(1+u_x), \frac{1}{p+4}(-(3p+4)tv_t + pv) - xv_x, \frac{1}{p+4}(-(3p+4)tw_t + pw) - xw_x)$ gives

$$T_{12} = \frac{3p+4}{p+4}tH + \rho A(x(u_t(1+u_x) + v_tv_x + w_tw_x) - \frac{p}{p+4}((x+u)u_t + vv_t + ww_t)),$$

$$X_{12} = \rho A(-\frac{1}{2}x(u_t^2 + v_t^2 + w_t^2) - \frac{p+1}{p+2}k_0x(1+\lambda)^{p+2} + \frac{1}{p+4}k_0(1+\lambda)^p(-(3p+4)t(u_t(1+u_x) + v_tv_x + w_tw_x) + p((x+u)(1+u_x) + vv_x + ww_x)))$$

- Case 3: $k(\lambda) = k_0(1 + \lambda)^{-4}$, $k_0 = \text{const.} \neq 0$

In this case, corresponding to the 13 variational symmetries, there are 13 conservation laws, including the specializations of the 11 from Case 1 plus two additional ones:

- i. scaling: $\vec{P} = (\frac{1}{2}(x + u) - tu_t, \frac{1}{2}v - tv_t, \frac{1}{2}w - tw_t)$ gives

$$T_{13} = tH - \frac{1}{2}\rho A((x + u)u_t + vv_t + ww_t),$$

$$X_{13} = \rho A k_0 (1 + \lambda)^{-4} (-t(u_t(1 + u_x) + v_tv_x + w_tw_x) + \frac{1}{2}((x + u)(1 + u_x) + vv_x + ww_x))$$

- ii. time-dependent (conformal) scaling: $\vec{P} = (t(x + u) - t^2u_t, tv - t^2v_t, tw - t^2w_t)$ gives

$$T_{14} = t^2H + \rho A(-t((x + u)u_t + vv_t + ww_t) + \frac{1}{2}((x + u)^2 + v^2 + w^2)),$$

$$X_{14} = \rho A k_0 t (1 + \lambda)^{-4} (-t(u_t(1 + u_x) + v_tv_x + w_tw_x) + (x + u)(1 + u_x) + vv_x + ww_x)$$

Conclusions

- *Derivations of the equations of motion for an elastic string*

The string equations can be derived from three different perspectives: Newtonian mechanics, nonlinear elasticity, and continuum mechanics. Each approach is consistent with the others.

- *Symmetries*

We found a 12-dimensional symmetry group for the string equations when the stress-strain function is arbitrary. There is one extra scaling symmetry in the power case and two additional symmetries: the scaling and a time-dependent scaling, in the special power case.

- *Conservation laws*

Using Noether's theorem, we found the conservation laws admitted by the string equations. They include the familiar conserved quantities: energy, um , and angular momenta.

- *No travelling wave solutions*

A direct substitution of the travelling wave solution ansatz $\vec{r} = \vec{r}(x-ct)$ into the string equations, followed by integrating once, shows that travelling wave solutions do not exist.

- *Non-straight equilibrium configurations*

The equilibrium condition for an arbitrary string configuration with arc length parameter s , $\partial_s(T(s)\widehat{r}_s) = 0$, gives two solutions:

1. a straight line configuration with constant tension, or
2. an arbitrary curve configuration with zero tension.

- *Prospective areas of research:*

1. finding the symmetries and conservation laws in the case of a finite string with physical boundary conditions at the ends
2. generalizing the derivation of the string equations to membranes, the two-dimensional analog of strings.